Global Wealth University — SmartBio Project Defense Guidelines & Oral Playbook

Candidate: Ndukwe Benedict Chibuikem
Matriculation No: `GWU/BSC/CS/2020/36182`
Project Title: *Design and Implementation of a Cloud-Integrated Fingerprint Biometric Attendance and NUC Compliance Management System (Global Wealth University as Case Study)*
Supervisor: Engr. Ebofuai Joshua
Department: Computer Science & Software Engineering
Date of Defense: August, 2026

🎙 1. Structured 5–7 Minute Defense Presentation Script

Use this exact oral structure when introducing your project to the panel of examiners:

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┌──────────────────────────────────────────────────────────────────────
│                          ORAL DEFENSE TIMELINE & SCRIPT
├─────────┬─────────────────┬──────────────────────────────────────────
│ Min 1   │ Introduction    │ Problem statement & NUC 75%
│ Min 2   │ Literature Gap  │ The "Sweaty/Injured Finger L
│ Min 3   │ Architecture    │ Real-time Cloud Sync + WebAu
│ Min 4-6│ Live Demo       │ Lecturer Start → Kiosk Flag
│ Min 7   │ Conclusion      │ NDPA 2023 compliance & Acade
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🗣 Spoken Script (Word-for-Word Guide):

> *"Good morning, respected Chairman of the Defense Panel, my supervisor Engr. Ebofuai Joshua, and distinguished members of the Department of Computer Science and Software Engineering.*
>
> *My name is **Ndukwe Benedict Chibuikem**, matriculation number **GWU/BSC/CS/2020/36182**. Today, I present my final year project titled: **Design and Implementation of a Cloud-Integrated Fingerprint Biometric Attendance and NUC Compliance Management System**.*
>
> *In Nigerian and West African tertiary institutions, the National Universities Commission (NUC) mandates that students achieve a minimum of **75% lecture attendance** before being certified eligible for semester examinations.*
>
> *However, traditional attendance management relies on paper roll-calls, which suffer from **three critical failures**:*
>
> 1. *Pervasive proxy signing where absentees have friends forge signatures.*
> 2. *Loss and physical degradation of paper sheets during departmental transit.*
> 3. *Massive administrative calculation latency at the end of the semester.*
>
> *Furthermore, my critical review of 24 peer-reviewed empirical studies identified a **fundamental literature gap**: existing commercial and academic biometric systems suffer from the **'Sweaty or Injured Finger Lockout Failure'**. When a student arrives with sweaty or scarred fingers, optical sensors reject them, forcing lecturers to either halt the class or turn students away.*
>
> *To solve this, I designed and implemented **SmartBio**—a cloud-integrated Single-Page Application powered by Google Cloud Firestore for sub-200ms multi-device WebSocket synchronization, Firebase Auth with institutional passcode gatekeeping, browser-native WebAuthn FIDO2 biometrics, an automated NUC 75% clearance engine with anti-tamper QR dockets, and the **Flagged Exception***

> ***Workflow** complying with Nigeria Data Protection Act (NDPA) 2023.*
>
> *With your permission, I would like to demonstrate the live working system."*

💻 2. Step-by-Step Live Demonstration Script

Follow this sequence during your live demo to keep examiners engaged:

Step 1: Role-Based Gatekeeping & Authentication

- **What to Show**: Click `📱 Switch User` → Select **`Register / Sign Up`**.
- **What to Explain**:

> *"To prevent students from creating unauthorized lecturer or administrative accounts, SmartBio implements **Institutional Passcode Gatekeeping**. Selecting Lecturer or Class Rep requires institutional authorization keys (e.g. `GWU-FACULTY-2026`)."*

- **Action**: Sign in as **Dr. Olawale Adeyemi** (`o.adeyemi@smartbio.edu.ng` / `password123`).
- **Highlight**: Notice how the navigation bar automatically locks down, hiding student and admin portals to enforce the **Principle of Least Privilege (PoLP)**.

Step 2: Lecturer Starts Live Session

- **What to Show**: In Lecturer Portal, select course `CSC 401: Advanced Software Architecture`, topic *"Distributed WebAuthn Biometrics"*, venue *"ICT Hall A"*, and click **`▶ Start Live Session`**.
- **What to Explain**:

> *"The moment the lecturer starts class, the session metadata and live timer are broadcast via Cloud Firestore WebSockets to all kiosks and students across campus in real time."*

Step 3: Edge Case Simulation at Kiosk Scanner

- **What to Show**: Switch to `🖲 Scanner Terminal`.
- **Action**: Select student **Tunde Bakare**, choose `Sweaty / Blurred Ridge`, and click `Simulate Optical Finger Scan`.
- **What Happens**: The scanner plays an acoustic warning double-beep, and HUD shows `FLAGGED EXCEPTION: SWEATY_RIDGE_BLURRED`.
- **What to Explain**:

> *"Instead of disrupting class or barring the student, SmartBio routes degraded scans into a real-time Flagged Queue on the lecturer's radar while the student takes their seat."*

Step 4: Lecturer Overrides Flag with NDPA 2023 Audit Trail

- **What to Show**: Switch back to `🧑‍🏫 Lecturer Portal`.
- **Action**: See Tunde Bakare appear in the Flagged Queue. Click `Review & Override`, enter remark: *"Verified student physical University ID card"*, and click `Approve & Credit Attendance`.
- **What to Explain**:

> *"The student receives their attendance credit, and an immutable audit log is generated containing the lecturer's ID, timestamp, and verification justification, satisfying NDPA 2023 compliance standards."*

Step 5: Student NUC 75% Compliance & Anti-Tamper QR Docket

- **What to Show**: Switch to `🎓 Student Portal` (`b.uche@student.gwu.edu`).

- **What to Explain**:

> *"SmartBio calculates the exact statutory percentage across all registered courses. For students below 75%, it calculates an automated deficit forecast ($x = \lceil(0.75C - A)/0.25\rceil$) showing exactly how many upcoming classes they must attend to regain clearance."*

- **Action**: Click `📜 Generate Exam Clearance Docket`.
- **Highlight**: Show the official clearance docket with the student's photo, status stamps, and cryptographically verifiable QR verification token.

🎯 3. Anticipated Difficult Examiner Questions & Winning Answers

#	Anticipated Examiner Question	Flawless Technical Answer
Q1	*"Why did you use web-based WebAuthn and optical simulation instead of physical Arduino fingerprint sensors?"*	**Answer**: *"Modern industry standard biometric architecture is shifting to zero-trust web authenticators (FIDO2/WebAuthn) already embedded in mobile devices and laptops. Physical USB/Arduino modules create driver dependency bottlenecks and single-point hardware lockouts. Our design supports both: WebAuthn for browser biometrics and standardized optical minutiae processing over WebSocket cloud bridges."*

Q2	*"How does your system comply with the Nigeria Data Protection Act (NDPA) 2023 regarding biometric privacy?"*	**Answer**: *"Under NDPA 2023 §30, storing raw biometric image files (like JPEG/BMP fingerprints) is a critical compliance violation. SmartBio adheres to the principle of Data Minimization: raw images are instantly processed into one-way SHA-256 minutiae coordinate hashes and discarded. Even if the database is breached, raw fingerprints cannot be reverse-engineered."*
Q3	*"What prevents a class representative or tech-savvy student from altering their attendance percentage?"*	**Answer**: *"We enforce strict Separation of Duties (SoD) and Route Guards. Class Reps can only proctor kiosks and view headcount; they cannot approve flags. At the database layer, Firestore Security Rules prohibit client-side modification of `/attendance_records` and `/audit_logs` without verified lecturer/admin JWT claims."*
Q4	*"What happens if there is an internet outage in the lecture hall during class?"*	**Answer**: *"SmartBio is built with an Offline-First cache. All scans are recorded locally into the browser's IndexedDB / LocalStorage data store with timestamps. Once network connectivity is restored, the CloudSync engine automatically pushes buffered records to Cloud Firestore."*
Q5	*"How does the mathematical deficit formula work for at-risk students?"*	**Answer**: *"Let C be total sessions conducted and A be sessions attended. To find additional consecutive classes x required to reach $75%$: $\frac{A + x}{C + x} \ge 0.75 \implies A + x \ge 0.75C + 0.75x \implies 0.25x \ge 0.75C - A \implies x = \left\lceil \frac{0.75C - A}{0.25} \right\rceil$. Our system calculates this dynamically."*

🏷️ 4. Demo Login Credentials Quick-Card (Keep on Desk)

DEFENSE CREDENTIALS QUICK REFERENC		
Role	Email	Passwor
Administrator	admin@smartbio.edu.ng	passwor
Lecturer	o.adeyemi@smartbio.edu.ng	passwor
Lecturer	n.okoro@smartbio.edu.ng	passwor
Class Rep	c.eze@student.gwu.edu	passwor
Student (80%)	b.uche@student.gwu.edu	passwor
Student (50%)	a.mohammed@student.gwu.edu	passwor